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A Guide to Attach GKE Clusters to Anthos

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By [Anil Kumar Y](#)

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Introduction

Anthos is a powerful hybrid, multi-cloud platform from Google Cloud that enables businesses to modernize their IT infrastructure and applications. In today's rapidly-evolving digital landscape, businesses must be agile, scalable, and secure to stay ahead of the competition. Anthos helps businesses achieve these goals by providing a consistent platform for managing and deploying applications across different environments, including on-premises data centers, public clouds, and edge locations.

Anthos is built on top of Google Kubernetes Engine (GKE), a managed Kubernetes service that provides a powerful platform for deploying and managing containerized applications. GKE enables businesses to easily create, scale, and manage Kubernetes clusters, simplifying deploying and managing applications in a containerized environment.

Pioneers in Cloud Consulting & Migration Services

- Reduced infrastructural costs
- Accelerated application deployment

Get Started

Benefits of Anthos

Hybrid and Multi-Cloud Management: Anthos provides a consistent platform for managing and deploying applications across different environments, simplifying the process of managing complex hybrid and multi-cloud environments. This enables businesses to maintain consistency and control across different environments, improving agility and reducing complexity.

Advanced Security and Compliance: Anthos provides a range of advanced security and compliance features, such as encryption at rest and in transit, identity and access management, and compliance monitoring. These features help businesses maintain security and compliance across different environments, ensuring their applications and data are protected from security threats and regulatory violations.

Increased Scalability and Agility: Anthos enables businesses to create, scale, and manage Kubernetes clusters easily, simplifying deploying and managing applications in a containerized environment. This improves scalability and agility, reducing the time it takes to deploy new applications and features.

Improved Collaboration and DevOps: Anthos provides a collaborative environment for developers, operations teams, and security teams, enabling them to work together more effectively. This helps to streamline the development and deployment process, reducing time to market and improving application quality.



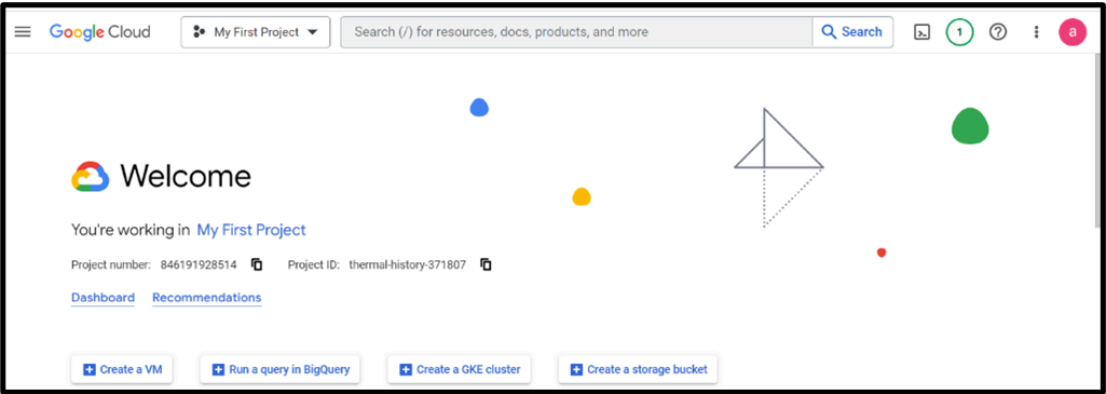


Step-by-Step Guide

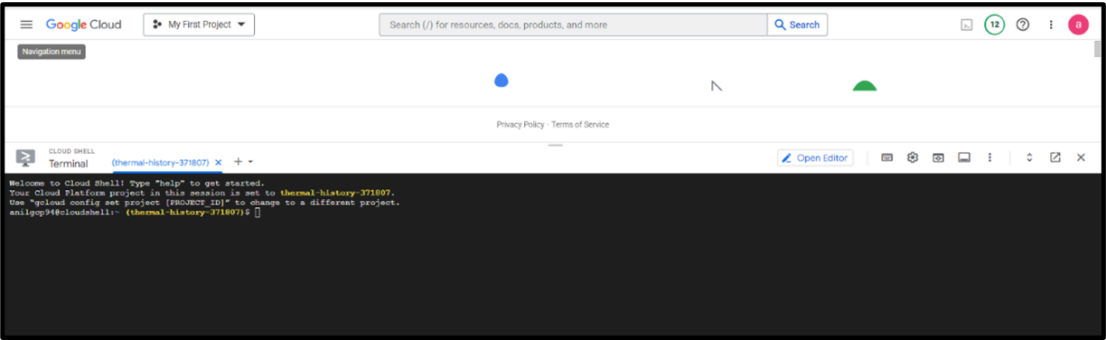
One of the key components of Anthos is Google Kubernetes Engine (GKE), which provides a managed Kubernetes service for deploying and managing containerized applications. This blog will provide a step-by-step guide for getting started with Anthos and attaching GKE clusters.

Step 1: Create 2 GKE clusters

- Login to the console, Select the project.



- Activate the cloud shell



- Create a sample Kubernetes cluster using the command:

```
1 $$ gcloud container clusters create cluster1 --machine-type n1-standard-2 --num-nodes 2 --zone us-central1-c

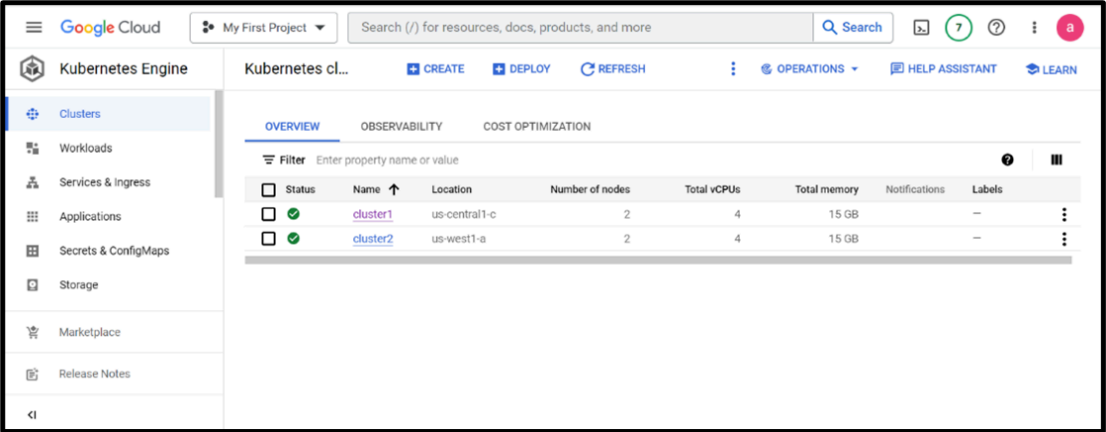
anilgcp94@cloudshell:~ (thermal-history-371807)$ gcloud container clusters create cluster1 --machine-type n1-standard-2 --num-nodes 2 --zone us-central1-c
Default change: VPC-native is the default mode during cluster creation for versions greater than 1.21.0-gke.1500. To create advanced routes based clusters, please pass the '--no-enable-ip-alias' flag
Default change: During creation of nodepools or autoscaling configuration changes for cluster versions greater than 1.24.1-gke.800 a default location policy is applied. For Spot and FVM it defaults to ANY, and for all other VM kinds a BALANCED policy is used. To change the default values use the '--location-policy' flag.
Note: Your Pod address range ('--cluster-ipv4-cidr') can accommodate at most 1008 node(s).
Creating cluster cluster1 in us-central1-c... Cluster is being health-checked (master is healthy)...done.
Created (https://container.googleapis.com/v1/projects/thermal-history-371807/zones/us-central1-c/clusters/cluster1).
To inspect the contents of your cluster, go to: https://console.cloud.google.com/kubernetes/workload/_gcloud/us-central1-c/cluster1?project=thermal-history-371807
kubeconfig entry generated for cluster1.
NAME: cluster1
LOCATION: us-central1-c
MASTER VERSION: 1.24.8-gke.2000
MASTER IP: 104.198.159.34
MACHINE TYPE: n1-standard-2
NODE VERSION: 1.24.8-gke.2000
NUM NODES: 2
STATUS: RUNNING

1 $$ kubectl get nodes

anilgcp94@cloudshell:~ (thermal-history-371807)$ kubectl get nodes
NAME                                STATUS    ROLES    AGE   VERSION
gke-cluster1-default-pool-ddd4418e-5qvb Ready    <none>   11m   v1.24.8-gke.2000
gke-cluster1-default-pool-ddd4418e-tc7d Ready    <none>   11m   v1.24.8-gke.2000
```

Repeat the same steps for creating another GKE cluster.

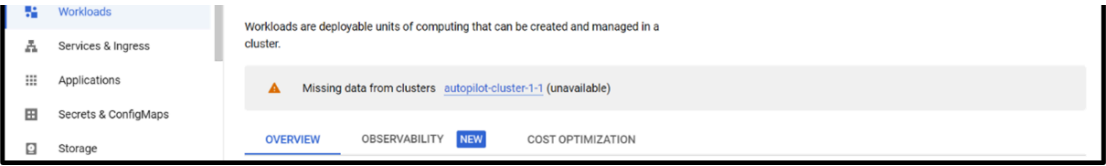
Once the clusters are ready, go to the console >> Kubernetes Engine >> Clusters and verify.



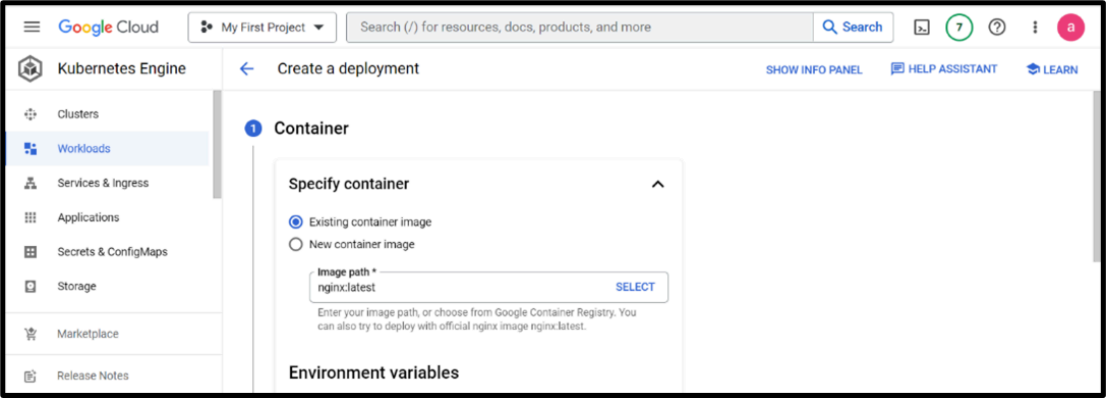
Step 2: Create a workload in the cluster.

- In Kubernetes Engine >> Workloads >> Deploy

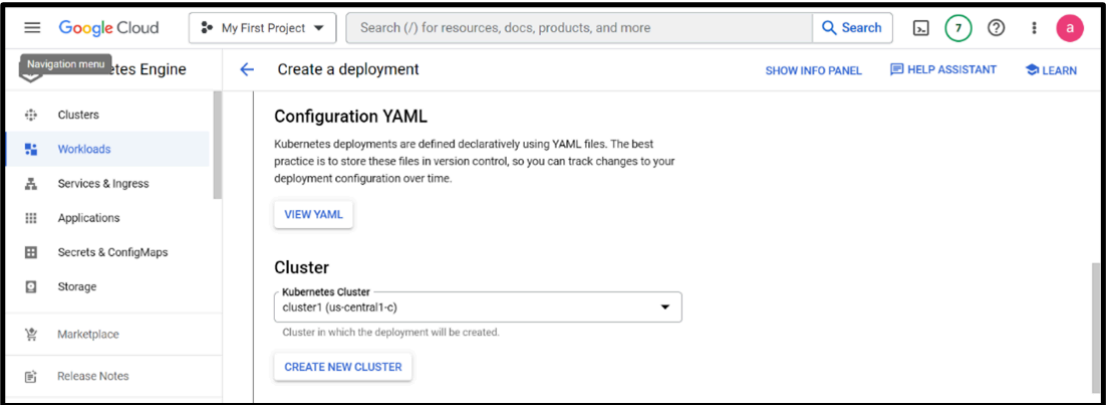
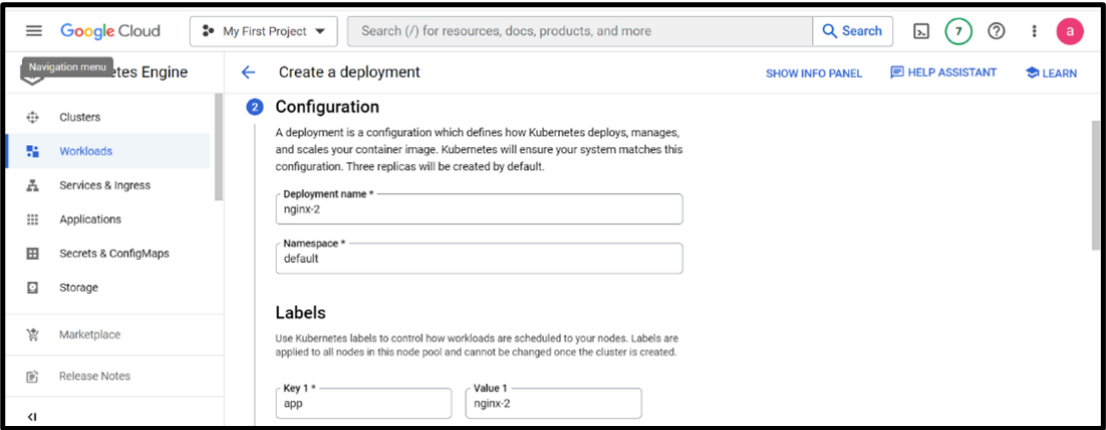




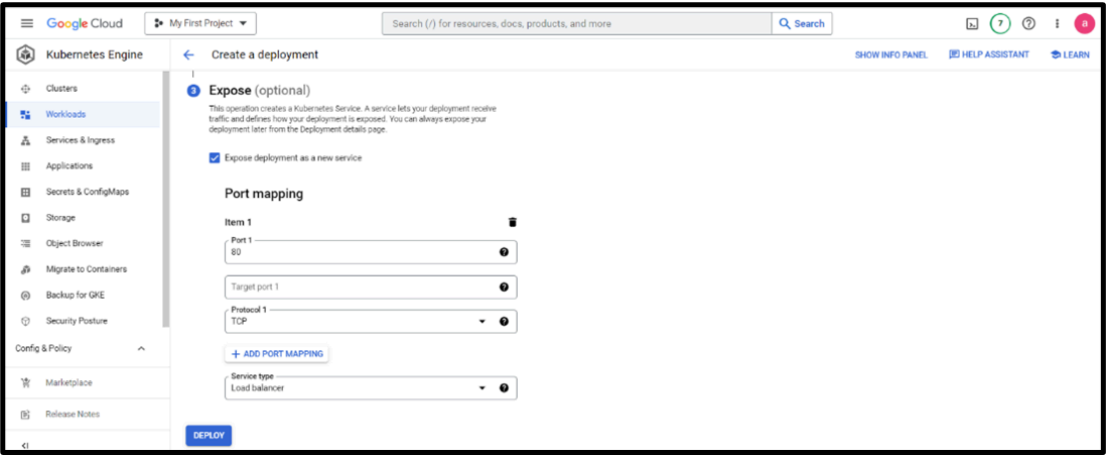
- Click on Deploy to create a deployment. Here select the existing container image and click on continue.



- In the configuration, enter the values for the deployment name, namespace, key any, and value of your choice. Select the cluster in which deployment is to be created.

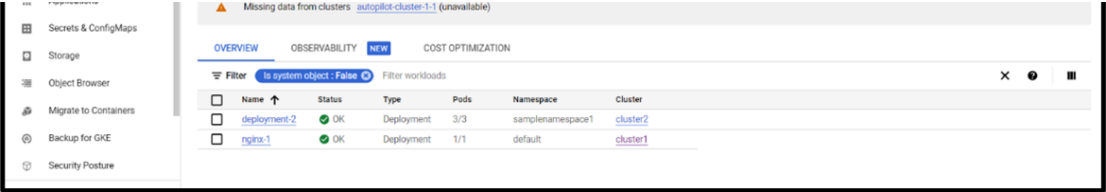


- Click on continue. Select expose deployment as a new service, enter the target port, and select the type of service.



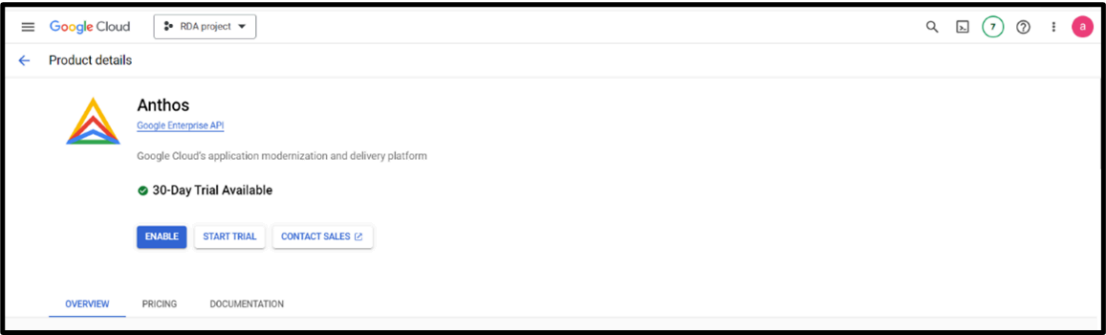
- Create the deployment for the other cluster in the same way.



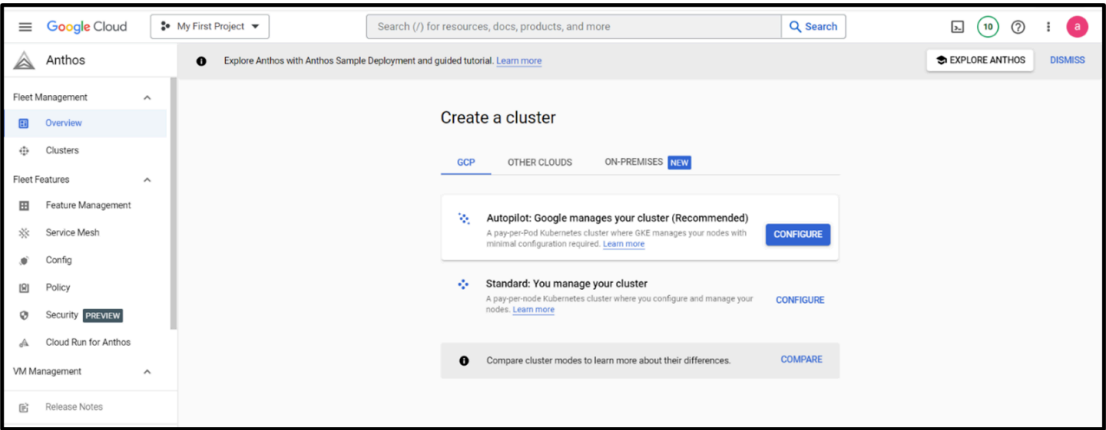


Step 3: Set up Anthos

- In the Google Cloud Console, navigate to the Anthos

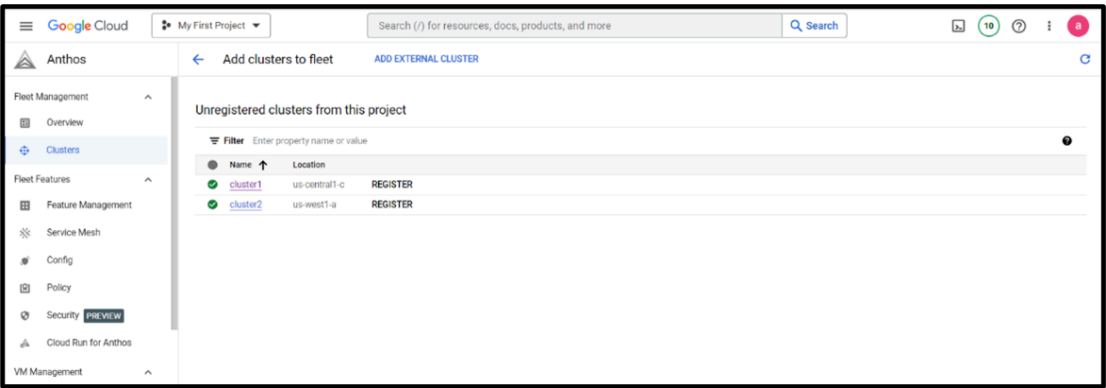
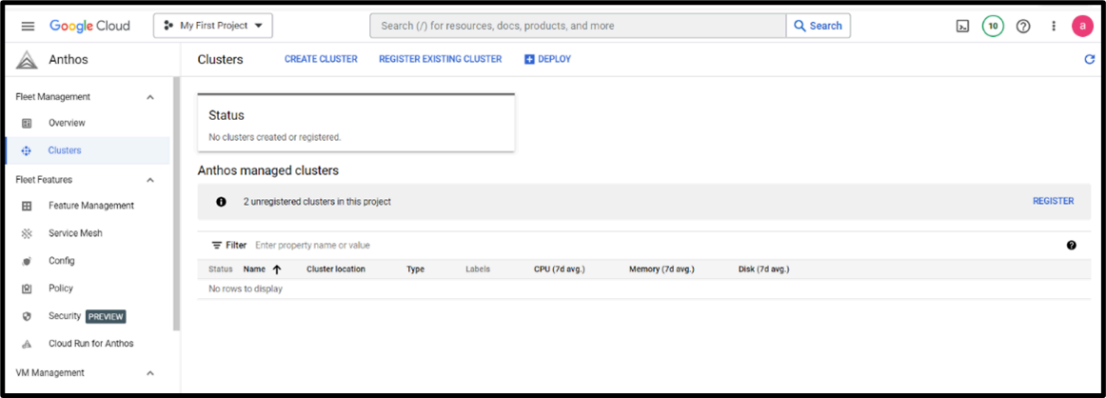


- Enable the Anthos, and click on the overview

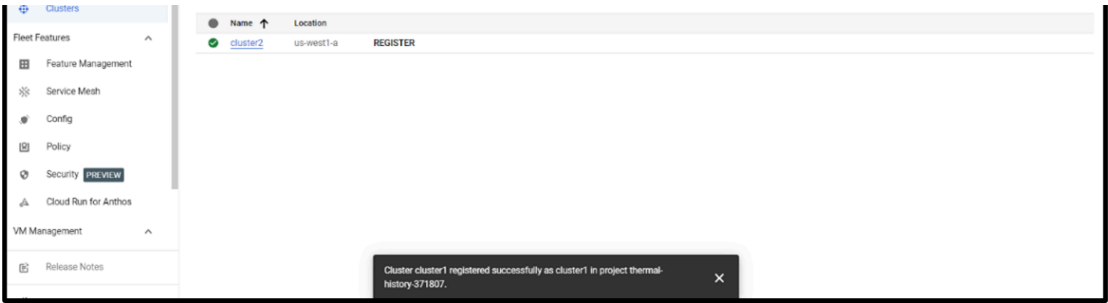


Step 4: Register the GKE cluster with Anthos

- In the Fleet management section of Anthos, choose the Clusters option, and then select the Register existing cluster button.

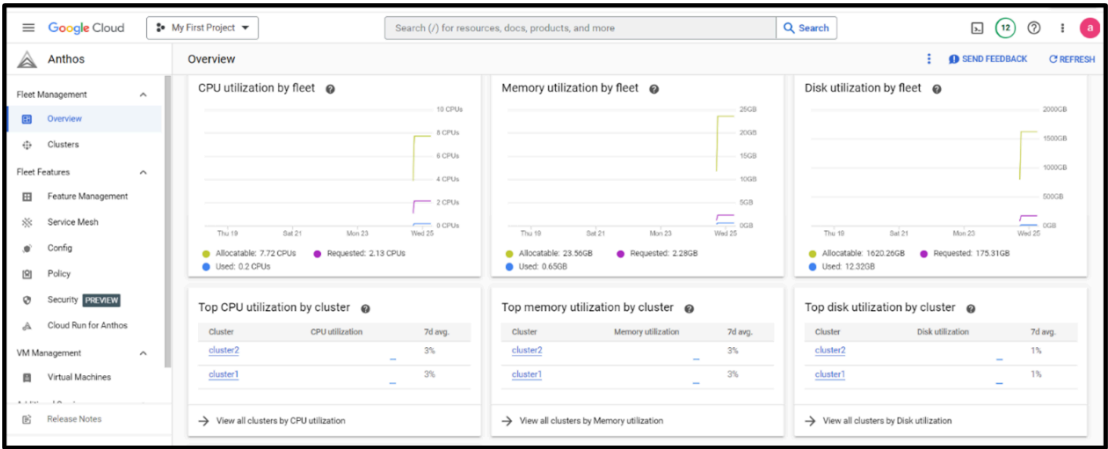


- To register with the fleet, click on the REGISTER button.



Step 5: Verify the GKE cluster is attached to Anthos

- Once the clusters get registered in the fleet, you can see the overview of the different parameters of the clusters.



Status	Name	Cluster location	Type	Labels	CPU (7d avg.)	Memory (7d avg.)	Disk (7d avg.)
Healthy	cluster1	us-central1-c	GKE		3%	3%	1%
Healthy	cluster2	us-west1-a	GKE		3%	3%	1%

Conclusion

This blog post provided a step-by-step guide for getting started with Anthos and attaching GKE clusters. By following these steps, you can easily attach your GKE clusters to Anthos and take advantage of the powerful hybrid and multi-cloud capabilities of Anthos. Whether modernizing your IT infrastructure or deploying new applications, Anthos provides a powerful platform for managing and deploying your applications across different environments.

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Google Cloud account, you can follow these steps to set up Anthos.

2. What other vendors support Anthos?

ANS: – Anthos is a hybrid and multi-cloud platform from Google Cloud that supports a variety of other cloud vendors, including Amazon Web Services (AWS) and Microsoft Azure.

3. Can Anthos speed up application modernization?

ANS: – Yes, Anthos can help speed up application modernization by providing a consistent platform for managing and deploying applications across different environments, including on-premises data centers, public clouds, and edge locations.

4. How does Anthos work across multiple environments?

ANS: – Anthos is designed to work across multiple environments, including on-premises data centers, public clouds, and edge locations. Anthos achieves this by providing a consistent platform for managing and deploying applications across different environments, enabling businesses to maintain consistency and control.

5. How does Anthos help secure my environment?

ANS: – Anthos provides a range of advanced security and compliance features to help secure your environment. At the core of Anthos security is Anthos Service Mesh, a unified platform for managing and securing microservices-based applications across different clusters and environments.

Anthos | Cloud Computing | GCP | GKE Clusters | Kubernetes |



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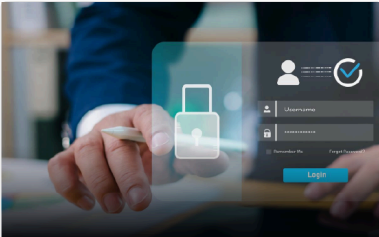
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